Financial Fraud Detection System

Who am I

- Background in Audio
- I work in Master control for a TV and Radio Station
- I work a lot with Music for Engineering and Live Sound
- Water connoisseur
- Massive Kansas City Chiefs fan
- The Favorite child

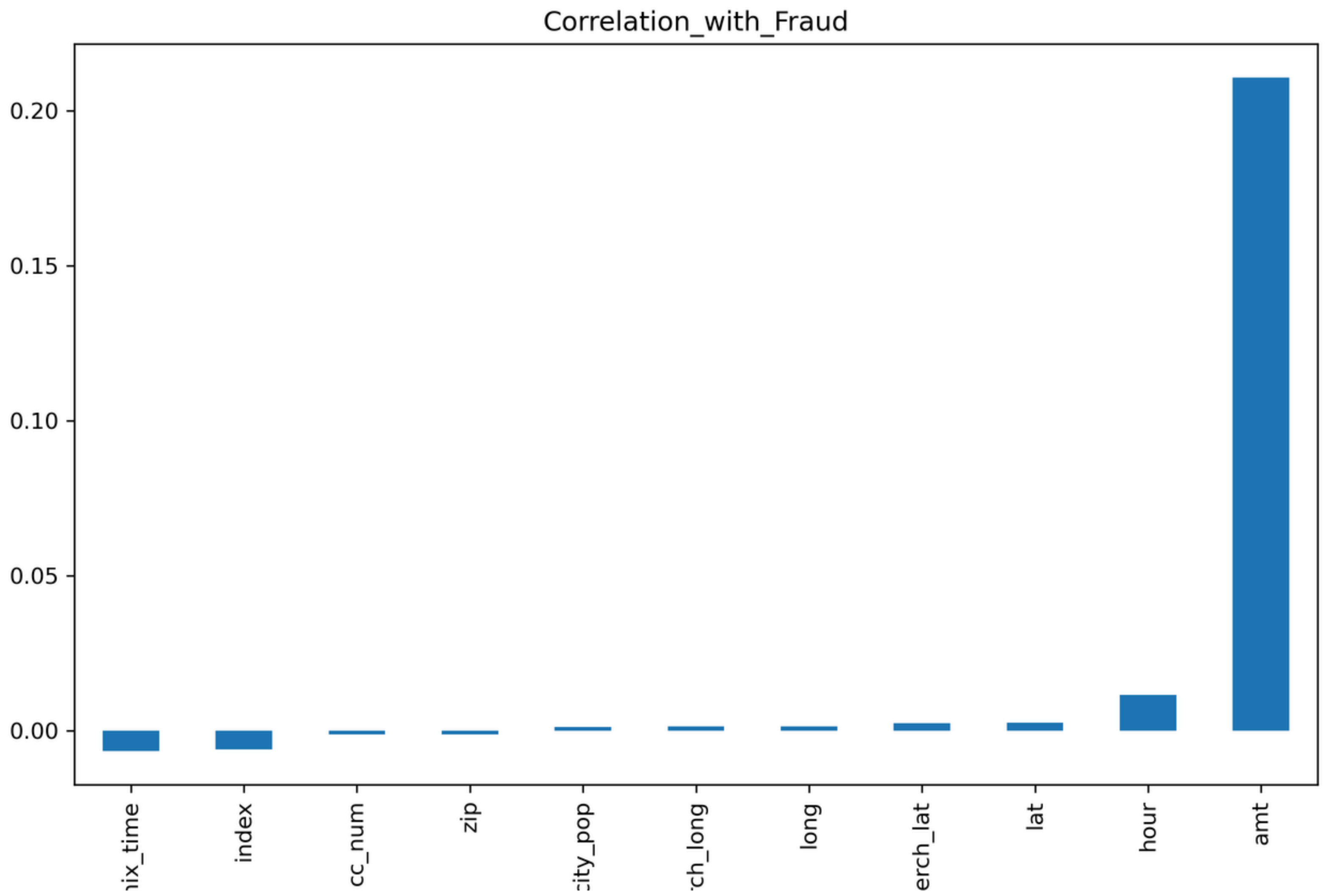
Dataset Overview and Methodology

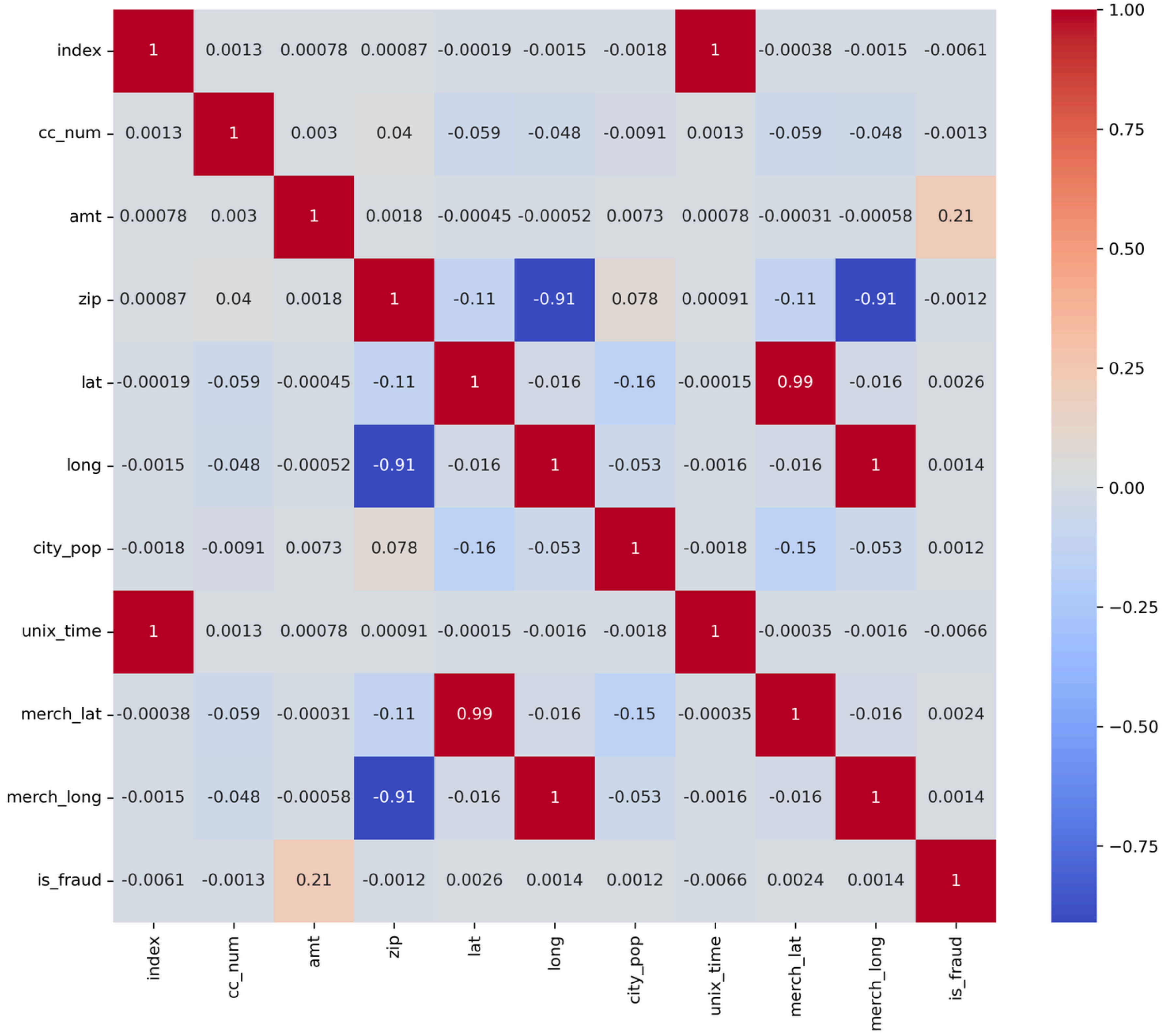
To gather and analyze the data for this project, I utilized a combination of Python, SQL, and Tableau. I used SQL to query, extract, and clean the data from the database, ensuring accuracy and consistency before any analysis was performed. Python was then used to conduct the analysis and create visuals, taking advantage of its versatile libraries to uncover patterns and present findings in a meaningful way. Finally, Tableau was used to build an interactive map, allowing for a clear and engaging geographic representation of the data.

Variable / Area	Key Finding
Total Records	389,002 transactions
Total Columns	23 (6 integer, 5 float, 12 object)
Missing Values	None across all columns
Duplicate Records	None detected
Fraud Rate	Approximately 0.58% (is_fraud mean = 0.005789)
Date Range	Full calendar year (January 1 to December 31)
Unique Cardholders	979 unique credit card numbers
Merchant Categories	14 distinct transaction categories

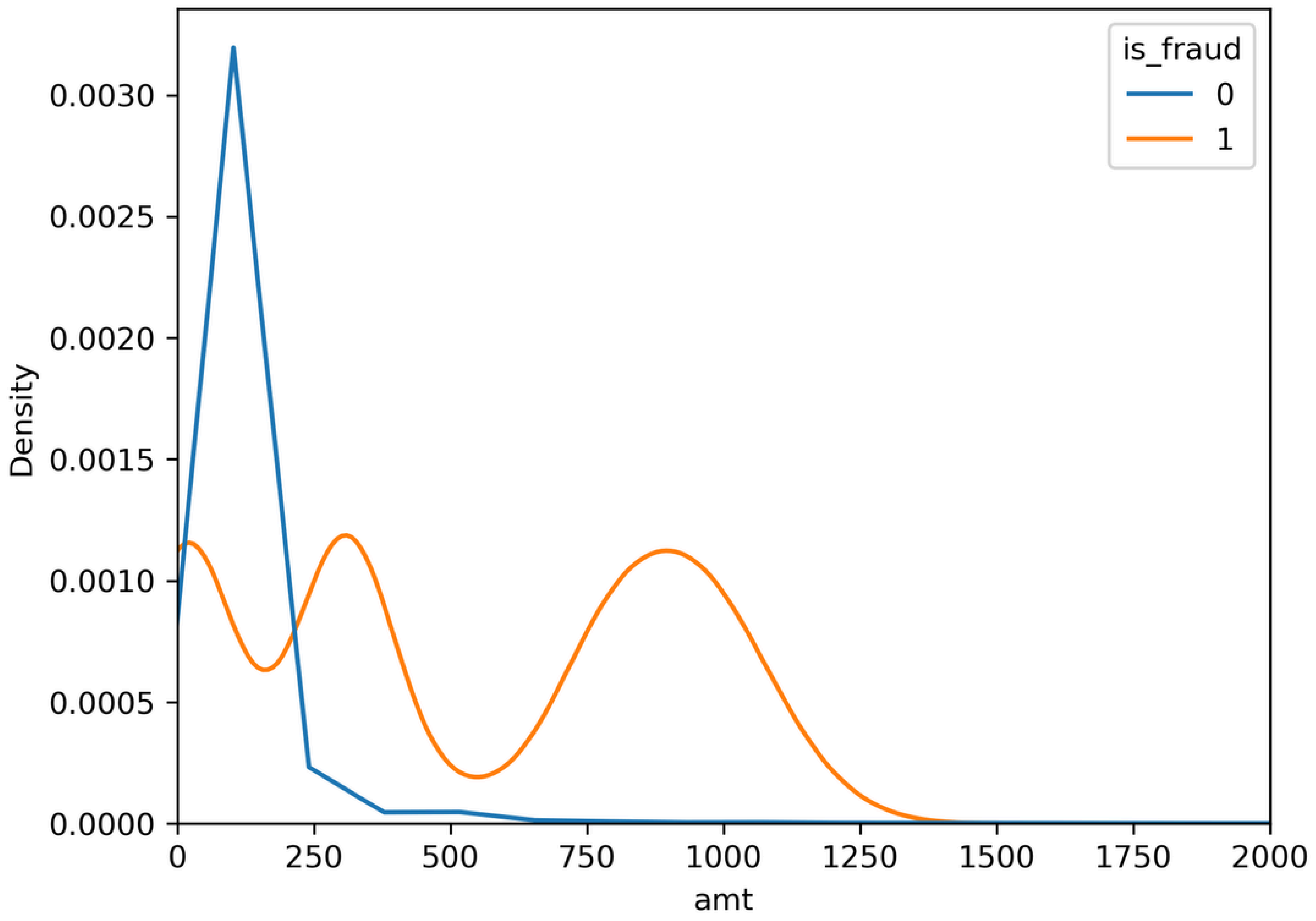
Correlation Analysis

- Transaction amount (amt) is the strongest numerical predictor of fraud (r = 0.21) All other variables show negligible correlation with fraud individually Geographic coordinates are highly correlated with each other but not with fraud



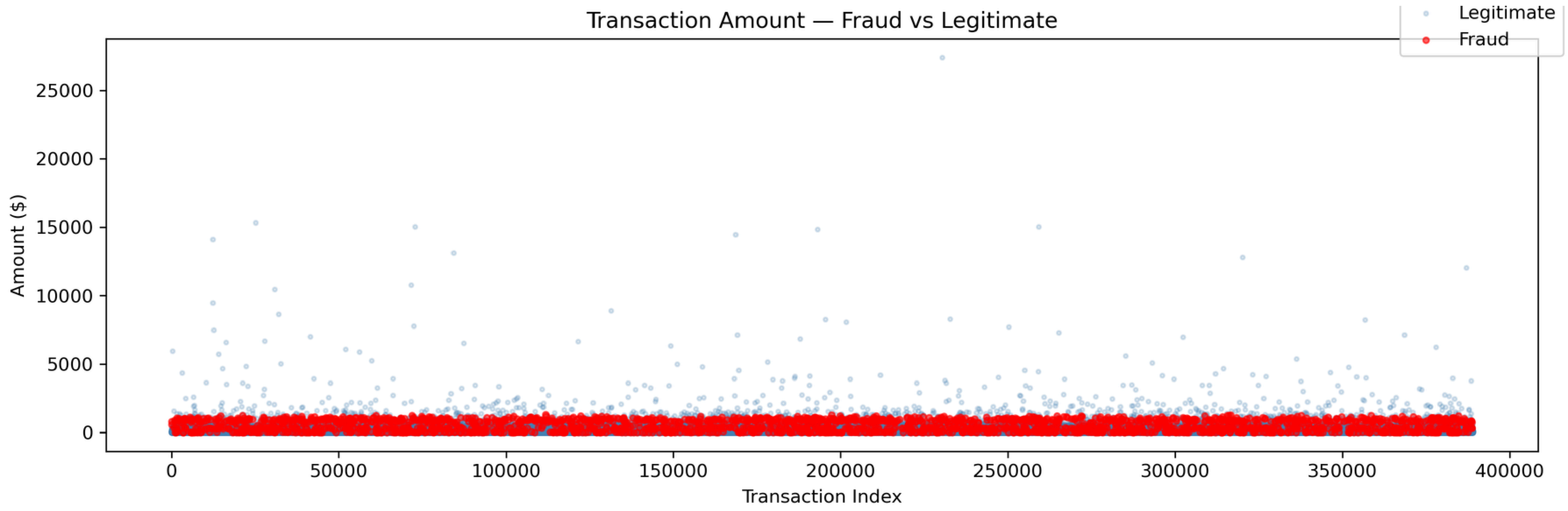


Transaction Amount Distribution



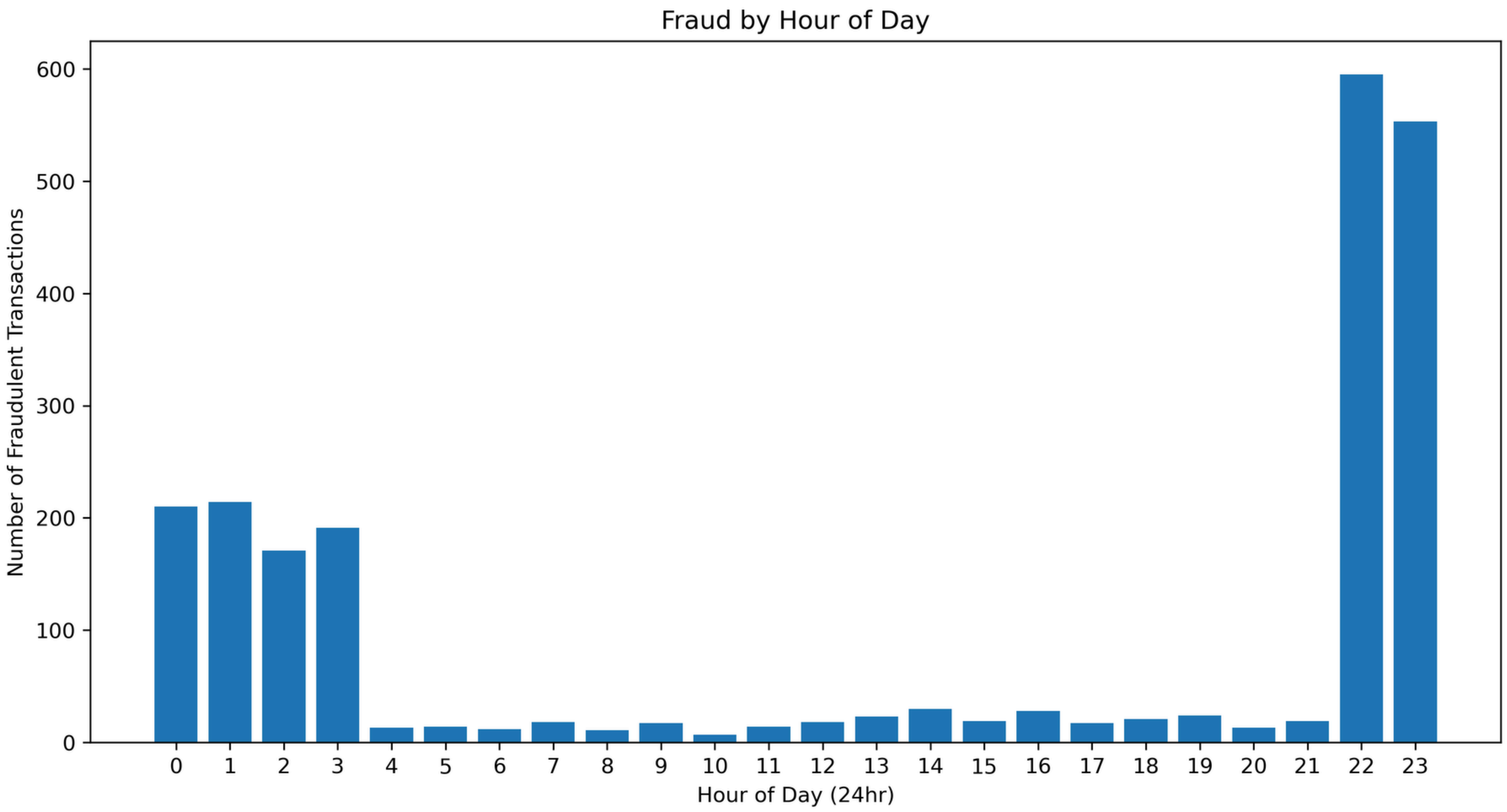
Legitimate transactions peak sharply below $100 and drop off quickly Fraudulent transactions show two distinct spending patterns — clustering around $250 and $900 Fraudsters avoid large obvious amounts, targeting mid-range purchases instead

Outlier Analysis



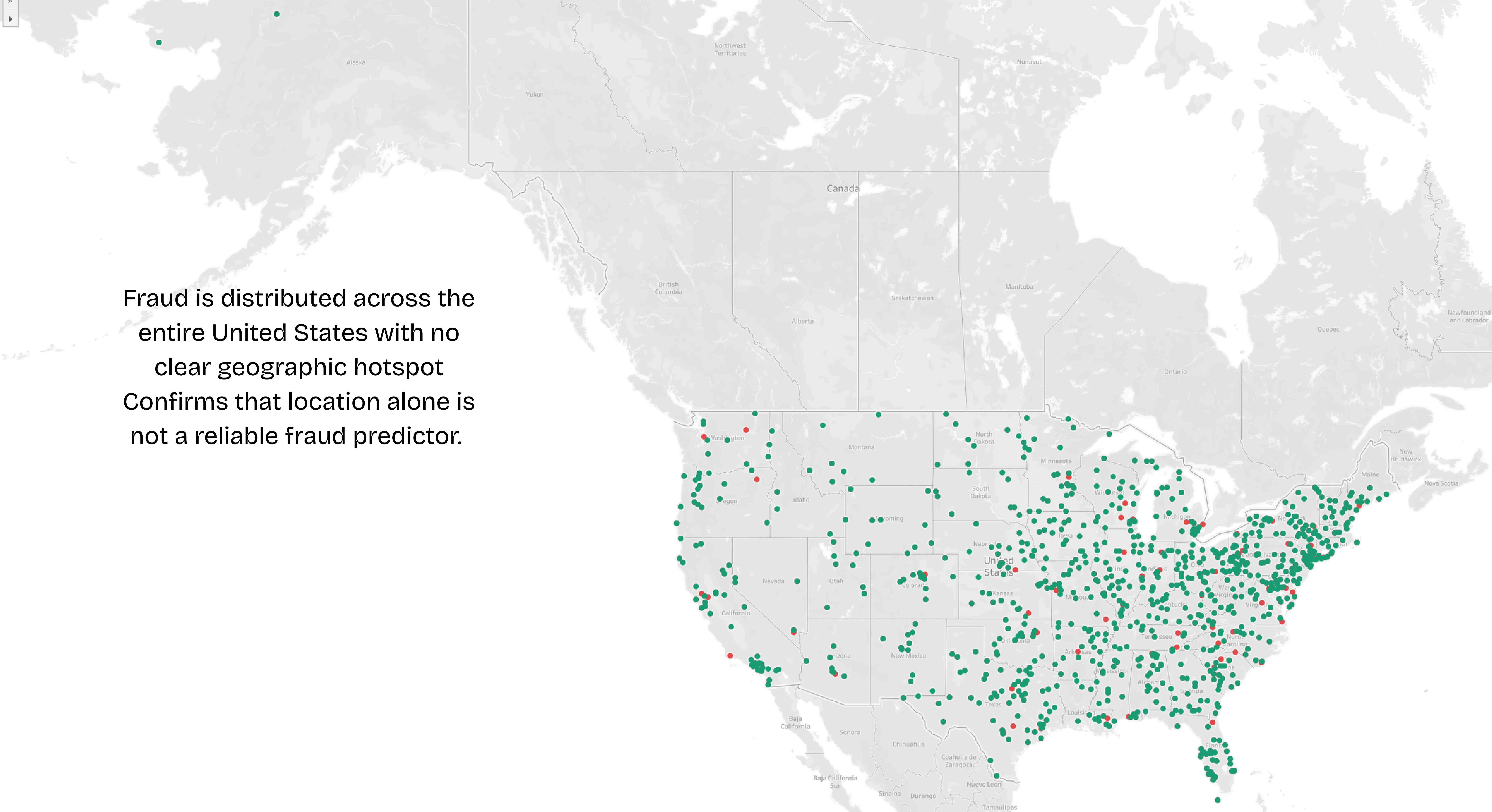
Median transaction amount: $47.57 Largest single transaction: $27,390.12 — flagged for investigation High-value outliers are predominantly legitimate, not fraud City populations range from small towns to 2.9 million — fraud occurs across all city sizes

Fraud by Hour



Fraud spikes dramatically between 10PM and 1AM Daytime hours (4AM–11AM) record fewer than 15 fraudulent transactions per hour Fraudsters operate when cardholders are asleep and less likely to notice

Fraud Location Map

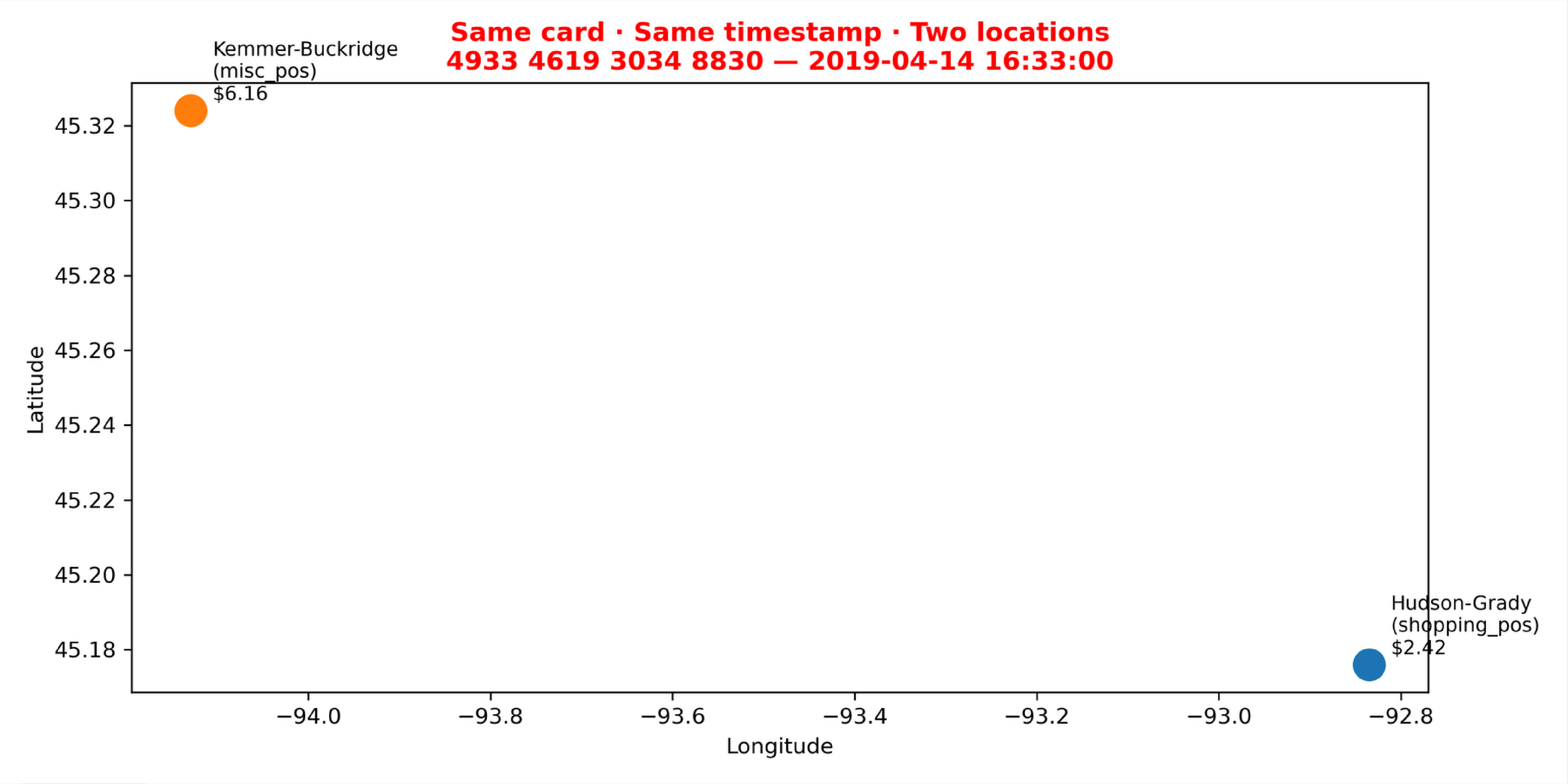


Fraud is distributed across the entire United States with no clear geographic hotspot Confirms that location alone is not a reliable fraud predictor.

Duplicate Transaction

3 key anomalies identified:

- Same card used at two different locations at the identical timestamp — physically impossible
- Cardholder aged 95 making a 4AM entertainment purchase — likely a data entry error
- 1,983 duplicate unix timestamps across the dataset warrant further investigation



Conclusions And Recommendations

- Transaction amount is the #1 fraud signal — weight it heavily in any model
- Increase monitoring sensitivity between 10PM and 1AM
- Class imbalance (0.58% fraud) must be addressed before model training using SMOTE or class weighting
- Fix mislabelled transactions before training to avoid noisy data
- Engineer a cardholder-to-merchant distance feature from the lat/long coordinates

Thank You!